

Problem Sheet 4

Learning on Topological Data:
Over-squashing and Over-smoothing

Topology and Geometry in Machine Learning

Lecture 4

Conventions. Throughout, $G = (V, E)$ denotes a finite, simple, undirected and connected graph, with at least two vertices, until self-loops are explicitly added. Let

$$A = (A_{uv}), \quad d_v = \sum_u A_{vu}, \quad D = \text{diag}(d_v), \quad L = D - A, \quad N = D^{-1/2} L D^{-1/2}.$$

Each sum over $\{u, v\} \in E$ counts an undirected edge once. For any node signal $f \in \mathbb{R}^{|V|}$, we denote its *Dirichlet Energy* as,

$$\mathcal{E}(f) := f^\top L f = \sum_{\{u, v\} \in E} (f_u - f_v)^2.$$

Remark. The exercises are marked in increasing difficulty by *, **, ***.

Exercise 1 (Normalized Laplacians) Add one unit self-loop at every vertex of $G = (V, E)$. This amounts to modifying the corresponding matrices as:

$$\tilde{A} = A + I, \quad \tilde{D} = D + I, \quad \tilde{L} = \tilde{D} - \tilde{A}, \quad \tilde{N} = \tilde{D}^{-1/2} \tilde{L} \tilde{D}^{-1/2}.$$

- (a, *) Show that $\tilde{L} = L$, but in general $\tilde{N} \neq N$. Prove that N and $\tilde{N}$ are symmetric. By substituting $f = D^{-1/2} z$, and then $f = \tilde{D}^{-1/2} z$, prove that both matrices are positive semidefinite.
- (b, **) Use $(a - b)^2 \leq 2(a^2 + b^2)$ to show

$$0 \leq z^\top N z \leq 2 \|z\|_2^2.$$

Deduce that every eigenvalue ν of N belongs to $[0, 2]$, i.e., $0 \leq \nu \leq 2$.

- (c, ***) Let $d_{\max} = \max_v d_v$. Prove the stronger augmented bound

$$0 \leq z^\top \tilde{N} z \leq \frac{2d_{\max}}{d_{\max} + 1} \|z\|_2^2.$$

Conclude that all eigenvalues $\tilde{\nu}$ of $\tilde{N}$ are strictly below 2, i.e., $0 \leq \tilde{\nu} < 2$.

- (d, *) Identify the kernels of N and $\tilde{N}$, using connectedness. Normalize their zero eigenvectors to unit length. Deduce that $P = I - \tilde{N}$ has a simple eigenvalue 1 and all other eigenvalues in $(-1, 1)$.
- (e, **) For the graph consisting of a single edge, compute N , $\tilde{N}$ and their spectra. Starting from $x = (1, 0)^\top$, compare $(I - N)^t x$ and $P^t x$. What obstruction to convergence have the self-loops removed?

Exercise 2 (Conductance, graph partitions and Cheeger inequalities) For a nonempty proper subset $S \subset V$, recall the definitions

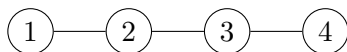
$$\partial S = \{\{u, v\} \in E : u \in S, v \notin S\}, \quad \text{vol}(S) = \sum_{v \in S} d_v, \quad \phi(S) = \frac{|\partial S|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}}.$$

Set $\phi_G = \min_S \phi(S)$. This exercise uses the original graph without self-loops, and ν_2 denotes the second-smallest eigenvalue of N .

Linear algebra tool. Changing variables $z = D^{1/2}f$ in the Rayleigh principle gives

$$\nu_2 = \min_{\substack{f \neq 0 \\ f^\top D \mathbf{1} = 0}} \frac{\mathcal{E}(f)}{f^\top D f}.$$

The orthogonality condition here is $\sum_v d_v f_v = 0$; for z it is $z \perp D^{1/2} \mathbf{1}$. Let G be the path on four vertices $1 - 2 - 3 - 4$:



(a, *) **Compute cuts.** Compute $\phi(S)$ for $S = \{1\}, \{2\}, \{1, 2\}, \{1, 3\}$ and $\{1, 4\}$. Use complementary sets and symmetry to determine ϕ_G . Why does minimizing the number of crossing edges alone not identify the best cut?

(b, *) **Run a spectral sweep.** You may use

$$\nu_2 = \frac{1}{2}, \quad Nu = \frac{1}{2}u, \quad u = (-\sqrt{2}, -1, 1, \sqrt{2})^\top.$$

Compute the scores $f = D^{-1/2}u$. Sort vertices by increasing f_v and evaluate the conductance of each nonempty proper prefix. Let γ be the smallest value. Check numerically that

$$\frac{\nu_2}{2} \leq \phi_G \leq \gamma \leq \sqrt{2\nu_2}.$$

(c, ***) **Prove the conductance lemma from the lecture.** For a general cut put $a = \text{vol}(S)$, $b = \text{vol}(V \setminus S)$ and $y = \mathbf{1}_S - \frac{a}{a+b} \mathbf{1}$. Show that

$$y^\top D \mathbf{1} = 0, \quad \mathcal{E}(y) = |\partial S|, \quad y^\top D y = \frac{ab}{a+b}.$$

Deduce the lemma and then the first Cheeger bound:

$$\nu_2 \leq \frac{\text{vol}(V) |\partial S|}{\text{vol}(S) \text{vol}(V \setminus S)}, \quad \frac{\nu_2}{2} \leq \phi_G.$$

Exercise 3 (Rewiring a bottleneck) Let G consist of two four-cycles

$$a_0 - a_1 - a_2 - a_3 - a_0, \quad b_0 - b_1 - b_2 - b_3 - b_0,$$

joined by the single edge $\{a_0, b_0\}$. Write $S = \{a_0, a_1, a_2, a_3\}$. You have a budget of two extra edges. Compare:

- **Across-community rewiring:** add $\{a_1, b_1\}$ and $\{a_3, b_3\}$, obtaining G_{cross} .
- **Within-community rewiring:** add $\{a_1, a_3\}$ and $\{b_1, b_3\}$, obtaining G_{inside} .

Compute conductances before adding self-loops.

- (a, *) Draw the three graphs. For each, compute $|\partial S|$, $\text{vol}(S)$, $\text{vol}(S^c)$ and $\phi(S)$. Which use of the edge budget improves this bottleneck? Explain why extra edges can also *decrease* the conductance of a fixed cut.
- (b, **) Compute the distance from a_2 to b_2 in each graph. What is the minimum number of local message-passing layers needed for a_2 to possibly depend on x_{b_2} ? Explain why shorter distance alone does not guarantee good predictions.
- (c, ***) For each graph form its own $P = (D + I)^{-1/2}(A + I)(D + I)^{-1/2}$ and propagate scalar features by $h^{(t)} = P^t x$. Show that

$$\frac{\partial h_{a_2}^{(t)}}{\partial x_{b_2}} = (P^t)_{a_2 b_2}.$$

At $t = 3$, prove that this derivative is zero for G and G_{inside} . For G_{cross} , enumerate all length-three walks from b_2 to a_2 and compute the derivative exactly. *Hint:* a walk contributes the product of its three propagation entries; self-loops cannot help a length-three walk if the distance is already three.

- (d, ***) Does improving $\phi(S)$ prove that the global conductance ϕ_G increases? Does adding edges increase every propagation entry? Compare $P_{a_1 a_2}$ in G and G_{cross} . Explain the possible tension between improved communication, normalization and over-smoothing.

Exercise 4 (Exact GCN limits and repeated input injection) Use the notations of Exercise 1 and additionally define

$$P = I - \tilde{N}, \quad Q = \sum_v (d_v + 1), \quad q = \frac{\tilde{D}^{1/2} \mathbf{1}}{\sqrt{Q}}.$$

We analyze the linear GCN from the slides: identity activation and $W^{(t)} = I$. Initially take scalar features $h^{(0)} = x \in \mathbb{R}^n$. Let $\tilde{N}\Phi_i = \tilde{\nu}_i\Phi_i$ be an orthonormal eigenbasis with $\Phi_1 = q$, and write $a_i = \Phi_i^\top x$, $\mu_i = 1 - \tilde{\nu}_i$.

- (a, *) **The usual propagation: exact nodewise limit.** For $h^{(t+1)} = Ph^{(t)}$, derive the finite-depth eigenbasis expression and prove

$$h^{(\infty)} = qq^\top x, \quad h_v^{(\infty)} = \frac{\sqrt{d_v + 1}}{Q} \sum_u \sqrt{d_u + 1} x_u.$$

Are these values equal at every node? What rescaling makes them equal? For the path 1–2–3 with $x = (1, 0, 0)^\top$, compute the three limiting values.

- (b, **) **Rate of collapse.** Let $\rho = \max_{i \geq 2} |\mu_i|$, from Exercise 1 we know that $\rho < 1$. Prove the following inequality:

$$\|h^{(t)} - qq^\top x\|_2 \leq \rho^t \|x - qq^\top x\|_2.$$

- (c, ***) **Injecting the initial features.** For $0 < \alpha < 1$ and $\beta = 1 - \alpha$, consider

$$h^{(t+1)} = \beta Ph^{(t)} + \alpha x, \quad h^{(0)} = x.$$

Write the update at a single node and expand the first two iterations.

Then prove, by induction, that

$$h^{(t)} = (\beta P)^t x + \alpha \sum_{k=0}^{t-1} (\beta P)^k x.$$

Derive the coefficient of each Φ_i at depth t , summing the geometric series explicitly.